

OHIDA BINTE AMIN

West Village Residence Complex H, 440 Huntington Ave, Boston, MA 02115

✉ amin.o@northeastern.edu 🌐 <https://ohidaamin.github.io/>

PROFESSIONAL SUMMARY

PhD Candidate in Personal Health Informatics with an interest in extending state-of-the-art ML innovations to improve healthcare systems by extracting better contextual & behavioral information from passive sensing data. Skilled in predictive analytics, statistical modeling, wearable sensing, and health informatics, with research and industry experience in assistive technologies. Experienced in multimodal modeling of behavioral and physiological data and translating research into scalable technologies that improve human performance.

EXPERIENCE

Khoury College of Computer Sciences, Northeastern University
Graduate Research Assistant (PhD Candidate) at Cognitive Systems Lab

September 2022 - Present

- Developing multimodal deep learning models using physiological, audio, and video data for generating continuous stress scores, achieving an AUC of 83.2%
- Deriving delirium biomarkers like respiration rates from sleep, activity, and physiological data collected from under-mattress sensors in older adults with dementia, achieving an AUC of 66.5% using anomaly detection models & mixed-effects modeling
- Building interpretable in-the-wild stress prediction models by integrating behavioral rules from passive sensing data and physiological biomarkers from a pre-trained SVM model from lab stressors with an AdaBoost model across temporal windows, achieving performance with AUC of 78.8%
- Conducted a controlled digital phenotyping stress study across research and consumer-grade wearables, applying SVM with RBF kernel and Random Forest, Garmin Forerunner 55s achieved highest performance, outperforming Empatica E4 by 6.2%

Samsung Research America
Data Science & Machine Learning Intern at Samsung Design Innovation Center

May 2024 – August 2024

- Designed methods to process and analyze unstructured text data, including feature extraction and sentiment analysis, to support dyslexia-related insights
- Developed deep learning models to predict personalized support needs and implemented a prototype system for dyslexia intervention

Baylor College of Medicine, Rice University
Research Fellow at The Fatima Al-Fihri Predoctoral Fellowship

April 2021 - December 2021

- Worked on automatic annotation of 3D genomic datasets using self-supervised deep learning approaches & conducted comparative study with different clustering techniques including K-means and DBSCAN on HiC features.

Robi Axiata Limited, Dhaka, Bangladesh
Data Science & Machine Learning Engineer

July 2020 - February 2022

- Developed a deep learning model as a complaint-classifier, handling training, validation, and deployment to resolve customer complaints with zero human intervention
- Built linear regression-based predictive model for suggesting suitable offers to the customers (like recharge, data pack, etc.) development, training, & validation
- Collaborated with cross-functional engineering teams and external partners to deploy new platform features, improve system performance, and address operational risks.

- Creating documentation for ML products and reviewing AI-assisted data labeling APIs & Abelling client data analysis by applying statistical analysis.

SKILLS

Language	Python, C, C++, R, Shell, Swift, SQL, Java
Frameworks/Tools	TensorFlow, PyTorch, Keras, Scikit-learn, PySpark, Pandas, NumPy, Matplotlib Oracle DBMS, MySQL, PyMongo, StatsModels, DEAP, Streamlit, Flask, Odoo, SkFuzzy, MATLAB Fuzzy Toolbox, KML, KML3D (R), OpenAI API
Domain	Machine Learning, Deep Learning, Digital Phenotyping, Digital Biomarkers, Mobile Sensing, Signal Processing, Data Science, Natural Language Processing, Human Computer Interaction
Wearable/Medical Devices	Empatica E4, Garmin, Polar Chest Straps, Biopac MP160 System

EDUCATION

Northeastern University, Boston, MA, USA	September 2022 - Present
PhD in Computer Science / Personal Health Informatics	
Northeastern University, Boston, MA, USA	April 2026
MS in Computer Science (not conferred)	
Khulna University of Engineering & Technology, Bangladesh	March 2020
BSc in Computer Science & Engineering	

RECENT PUBLICATIONS & PRESENTATIONS

[Google Scholar Profile](#)

-
- A. Sathyanarayana*, O.B. Amin*, J. An, and J.-P. Onnela, “*Examining the use of consumer wearable devices and digital tools for stress measurement in college students: a scoping review of methods*,” has been accepted for publication in JMIR mHealth and uHealth [\[Paper Link\]](#)
 - O.B. Amin, T.M. Tapera, R. Volpe, V. Mishra, and A. Sathyanarayana “*Extending Stress Detection Reproducibility to Consumer Wearable Sensors*,” in 2025 47th International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) **IEEE**, 2025. [\[Paper Link\]](#)
 - O.B. Amin, V. Mishra, and A. Sathyanarayana, “*Investigating Social Interaction Patterns with Depression Severity across Different Personality Traits Using Digital Phenotyping*,” in 2023 11th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW). **IEEE**, 2023. [\[Paper Link\]](#)
 - O.B. Amin, V. Mishra, and A. Sathyanarayana, “*The Impact of Stress and Sleep: Capturing Multi-day Patterns*,” in 2024 38th Annual Meeting of SLEEP, vol. 47, issue supplement1, 2024. [\[Abstract Link\]](#)

AWARDS & ACHIEVEMENTS

-
- SRS Diversity Membership Award 2023 – Sleep Research Society
 - Start-Up Fund Recipient – Northeastern University, Khoury College of Computer Sciences
 - Selected Research Fellow – The Fatima Al-Fihri Predoctoral Fellowship 2021

RELEVANT COURSES

Programming Design and Paradigm, Algorithms, Personal Health Interface Design and Development, Information Visualization Theory and Application, Mobile Application Development, AI for Human Computer Interaction, Understanding Users, Evaluating Health Technologies, Biostatistics in Public Health

TEACHING

CS 5200: Database Management Systems [Summer 2025 – Spring 2026]
PHTH 2210: Foundations of Biostatistics [Summer 2026]